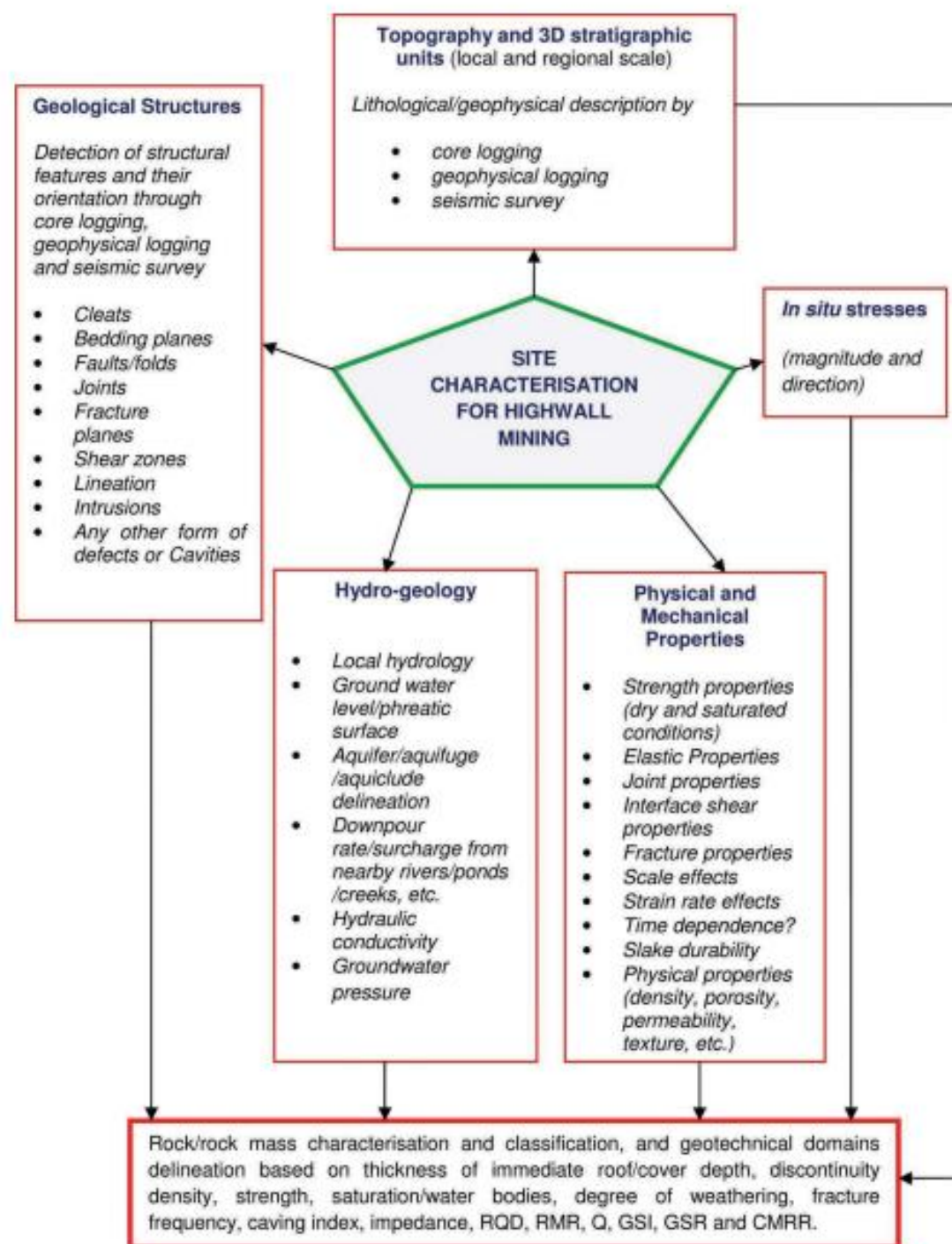
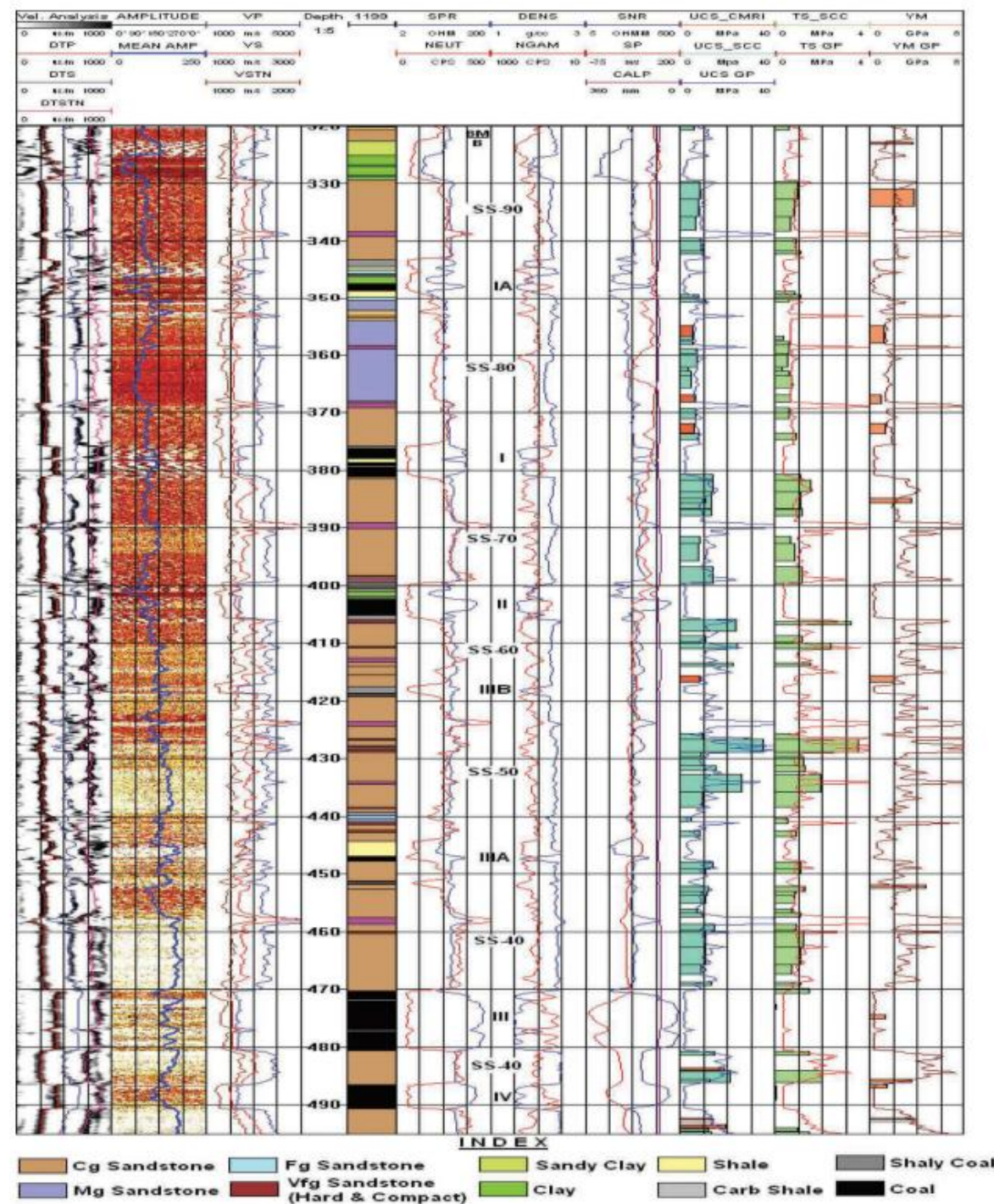
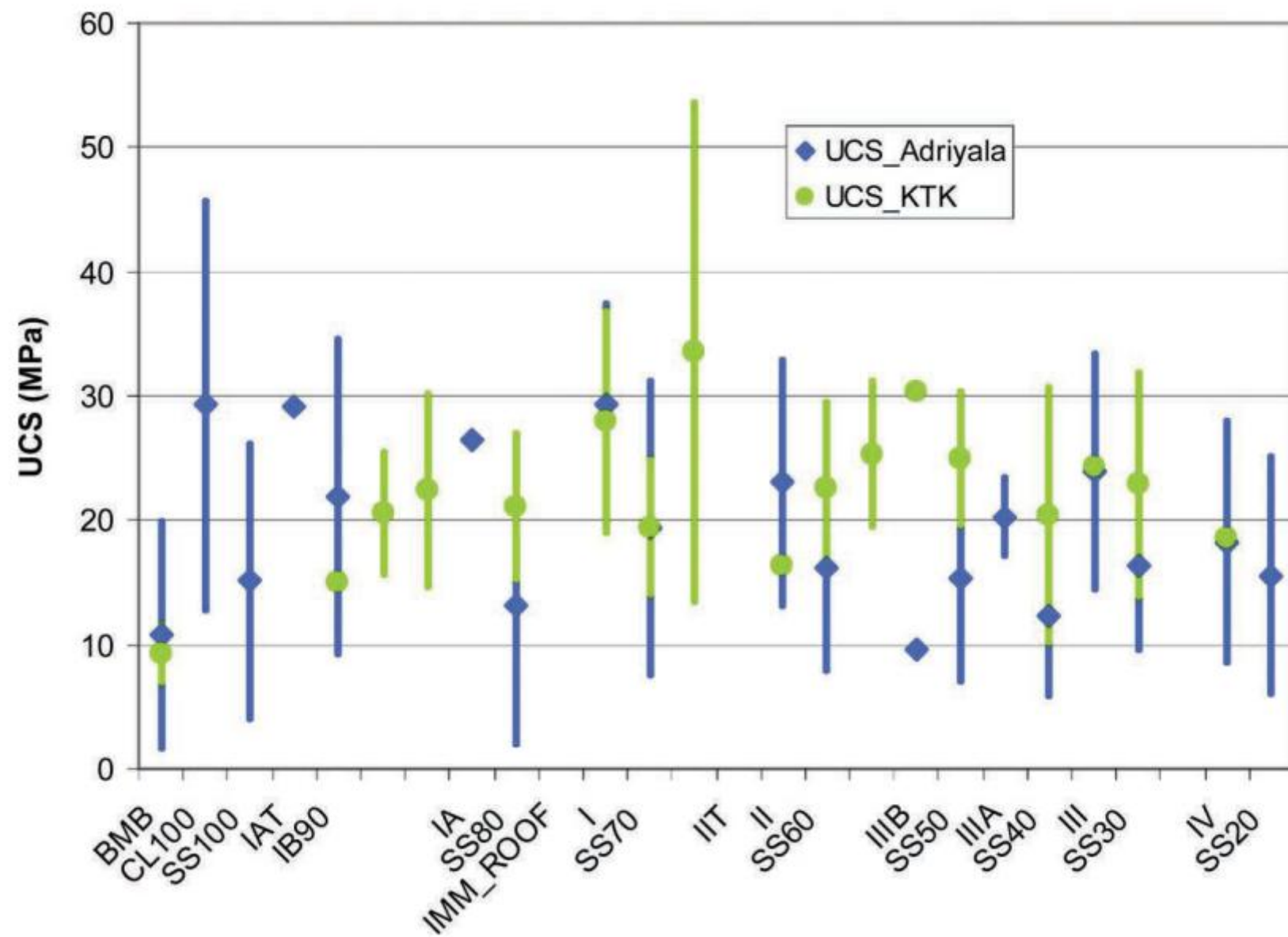


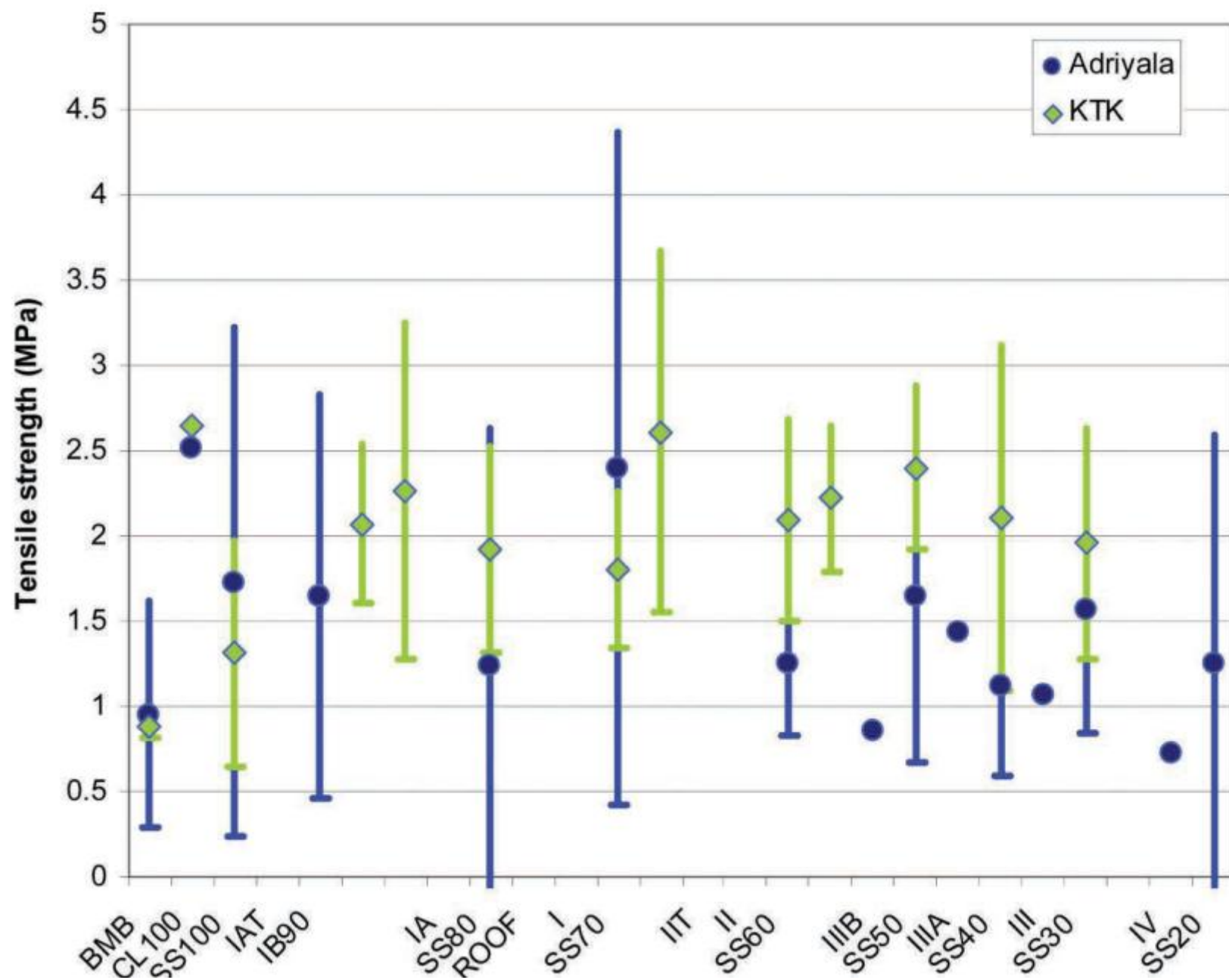
Geological Characterization of Highwall Mining Operations

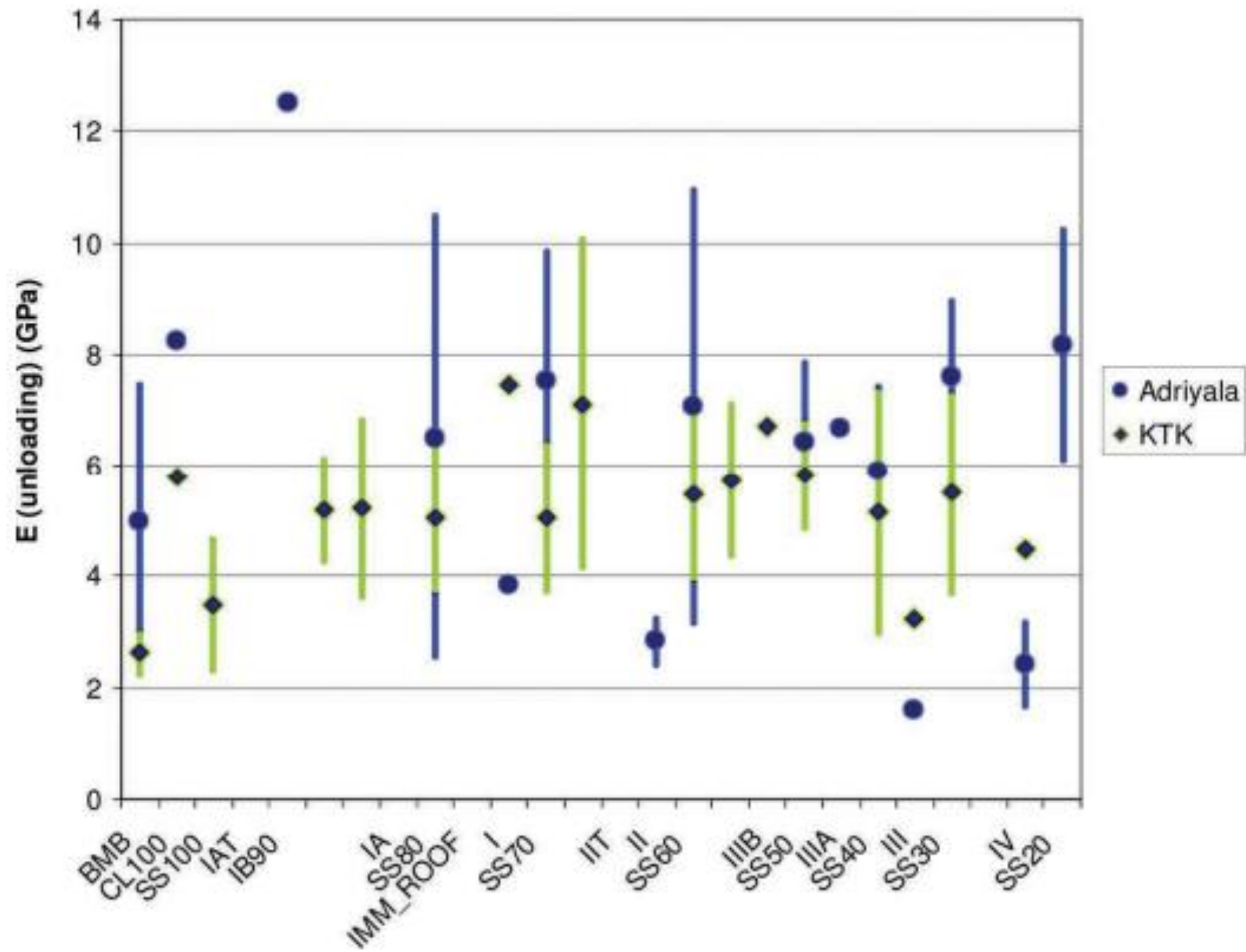


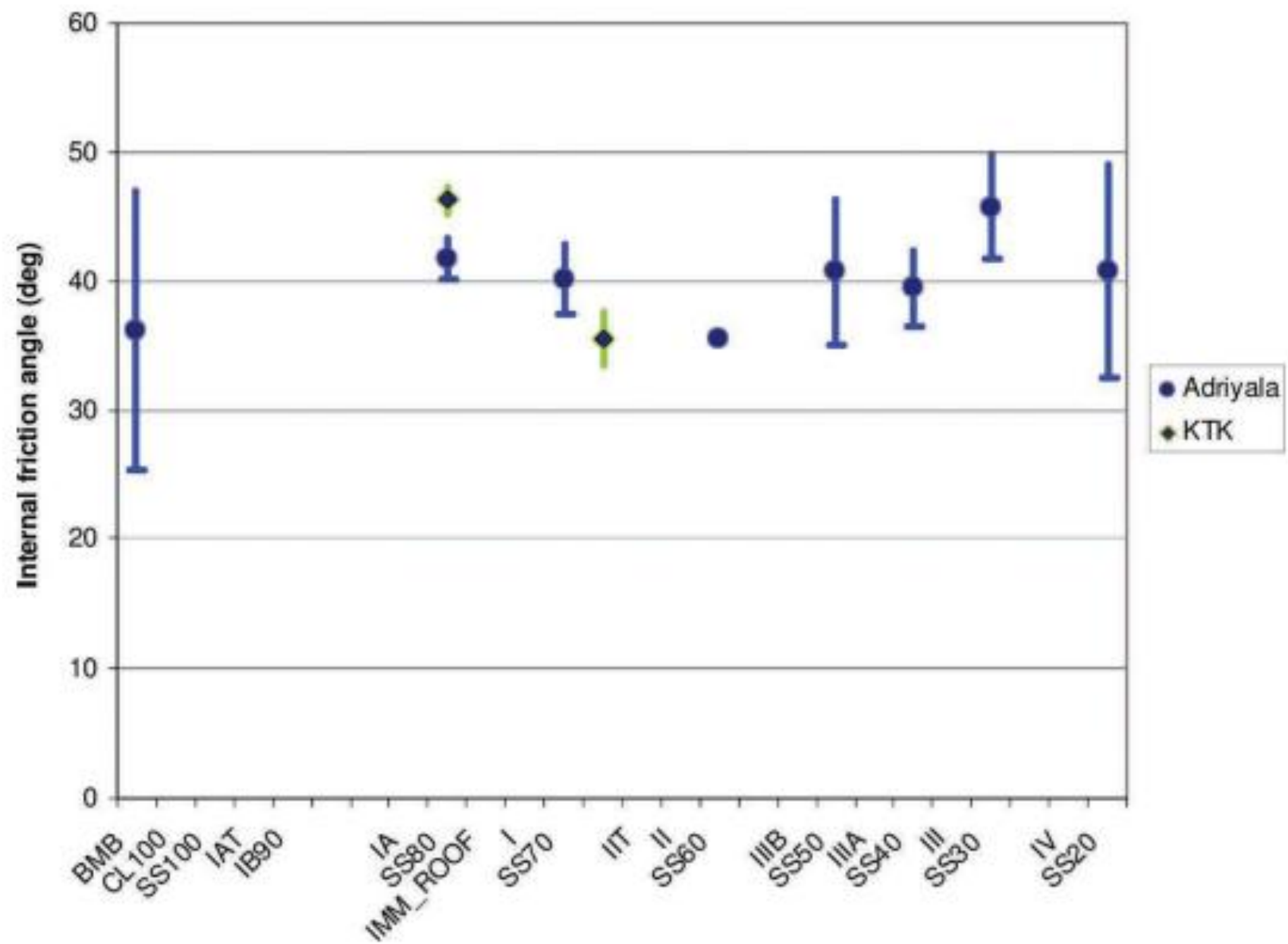


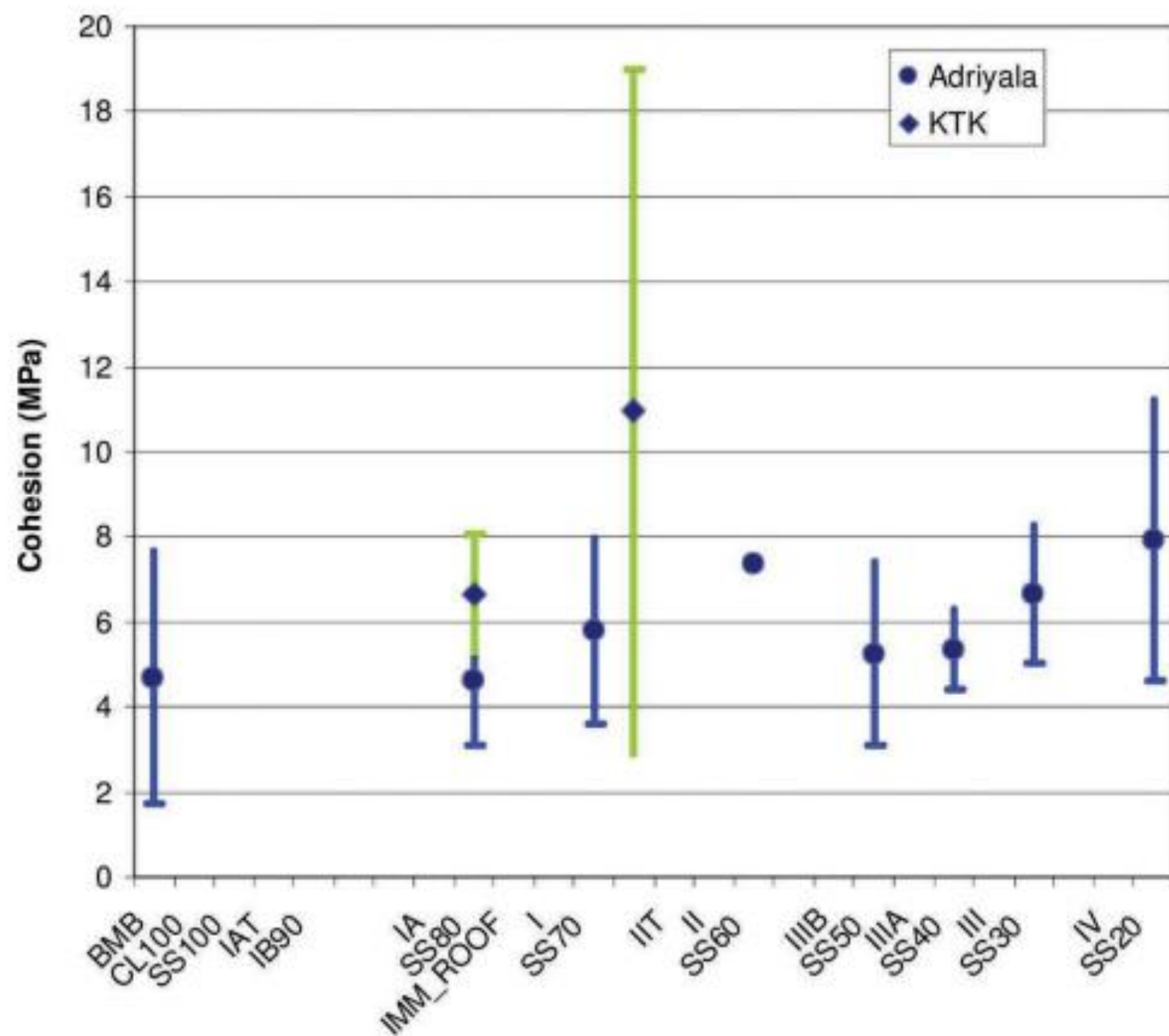




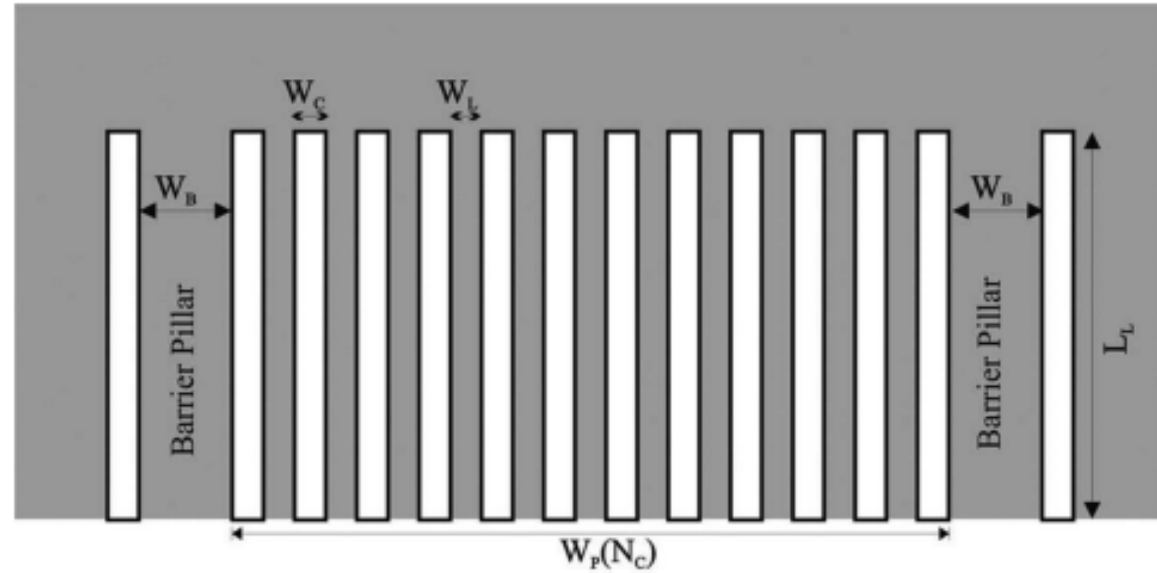








Design of Pillars



Coal Pillar Strength

- **Mark-Bieniawski formula** applies best for web pillars, which are very long, narrow rectangular pillars.

$$S_P = S_I [0.64 + 0.54 W / H]$$

Where:

S_P	=	web or barrier pillar strength
S_I	=	in situ coal strength
W	=	web or barrier pillar width
H	=	mining height

CSIR-CIMFR pillar strength formula

$$S = 0.27 * \sigma_c * h^{-0.36} + \left(\frac{H}{250} + 1 \right) \left(\frac{W_e}{b} - 1 \right)$$

where,

S is web pillar or barrier pillar strength (MPa)

H is the cover depth (m)

W_e is the equivalent pillar width (m) = $2W$ for long pillar

W is the width of the web pillar (m)

h is the working mining entry height (m)

σ_c is the strength of 25 mm cube coal sample (MPa)

Coal Pillar Stress

$$S_{WP} = S_V (W_{WP} + W_E) / W_{WP}$$

Where:

S_V	=	in situ vertical stress
W_{WP}	=	web pillar width
W_E	=	highwall miner hole width.

Depth Influence



$$D_{\text{Design}} = 0.75 * D_{\text{MAX}} + 0.25 * D_{\text{MIN}}$$

Factor of Safety / Stability Factor

$$SF_{WP} = \text{web pillar strength} / \text{web pillar stress}$$

Coal Pillar Stress for Barrier Pillar

$$S_{BP} = S_V (W_{PN} + W_{BP}) / W_{BP}$$

Where: W_{PN} = panel width
 W_{BP} = barrier pillar width

$$W_{PN} = N (W_{WP} + W_E) + W_E$$

Parameters	Value
Coal Seam Thickness, m	3
Highwall Miner Width, m	3.66
Maximum Depth of Cover, m	122,180,200
Minimum Depth of Cover (Highwall Height), m	35
Design Depth of Cover	?
Web Width, m	6,8,10
Number of Web Pillars	?
Panel Width	100m
Barrier Pillar Width, m	15,18,20